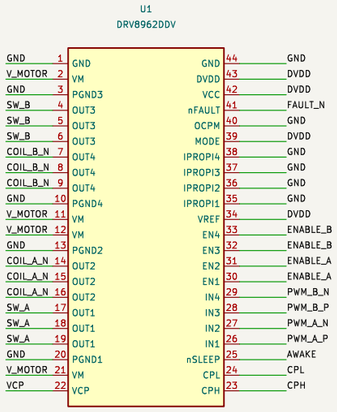
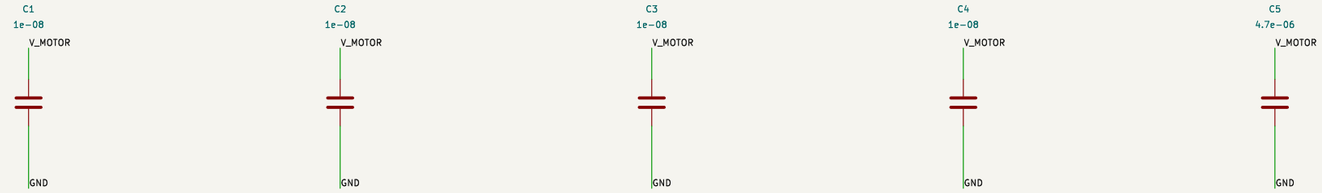
Integrated dual-coil circuit | DDV top-cooled package | DRAFT

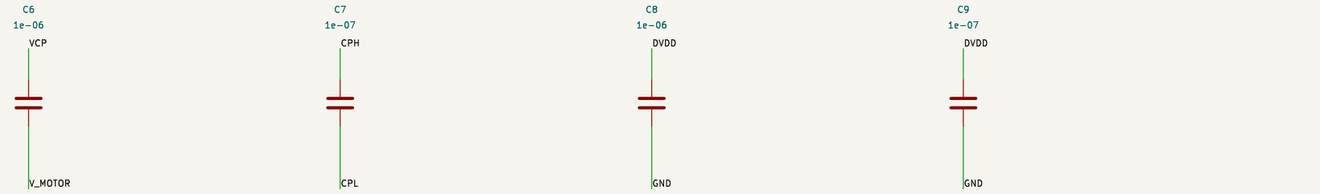
No SPICE model assigned. Shared supply, ADC/reference and control/protection sheets remain open.

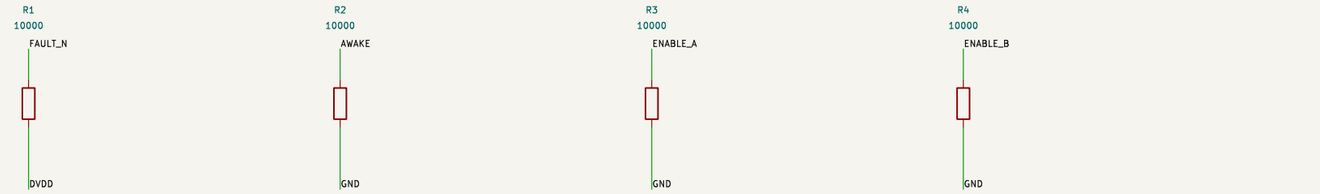
U1: DDV pin table; top thermal pad is NOT a bottom PCB pad.




Local VM bypass: one capacitor per VM pin; shared bulk/regen network still required.



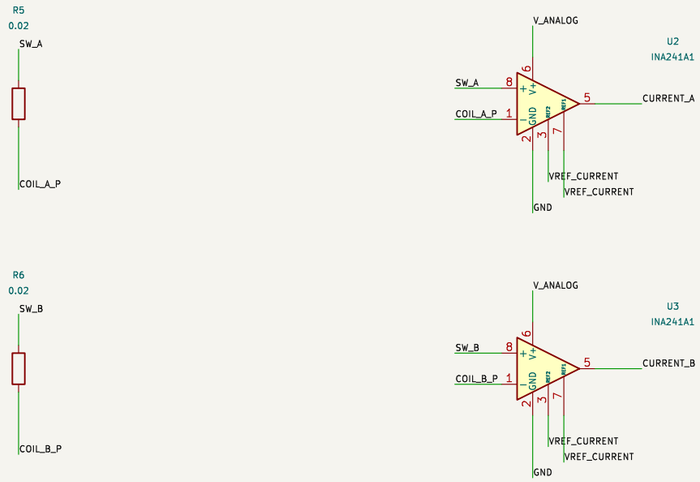




MODE high: fast edges. OCPM low: latched faults. IPROPI grounded: internal chopping disabled.

VREF and VCC use DVDD. Internal OCP is NOT the commanded-current or coil-safe current limit.

Inline Kelvin shunts measure both current polarities and recirculation. Coil path: P -> winding -> N.




External ports are circuit contracts, not connectors or completed shared circuitry.

Independent shutdown, ADC loading/filtering, timing, bus regeneration and cooling require verification.

V_MOTOR [input]

GND [input]

V_ANALOG [input]

VREF_CURRENT [input]

AWAKE [input]

ENABLE_A [input]

ENABLE_B [input]

PWM_A_P [input]

PWM_A_N [input]

PWM_B_P [input]

PWM_B_N [input]

FAULT_N [output]

CURRENT_A [output]

CURRENT_B [output]

COIL_A_P [bidirectional]

COIL_A_N [bidirectional]

COIL_B_P [bidirectional]

COIL_B_N [bidirectional]

Sheet: /
File: channel.kicad_sch

**Title: Integrated dual-coil driver and precision readout - DRAFT**

| Size: A3 | Date: | **Rev: draft** |
|---|---|---|
| KiCad E.D.A. 9.0.8 | | Id: 1/1 |